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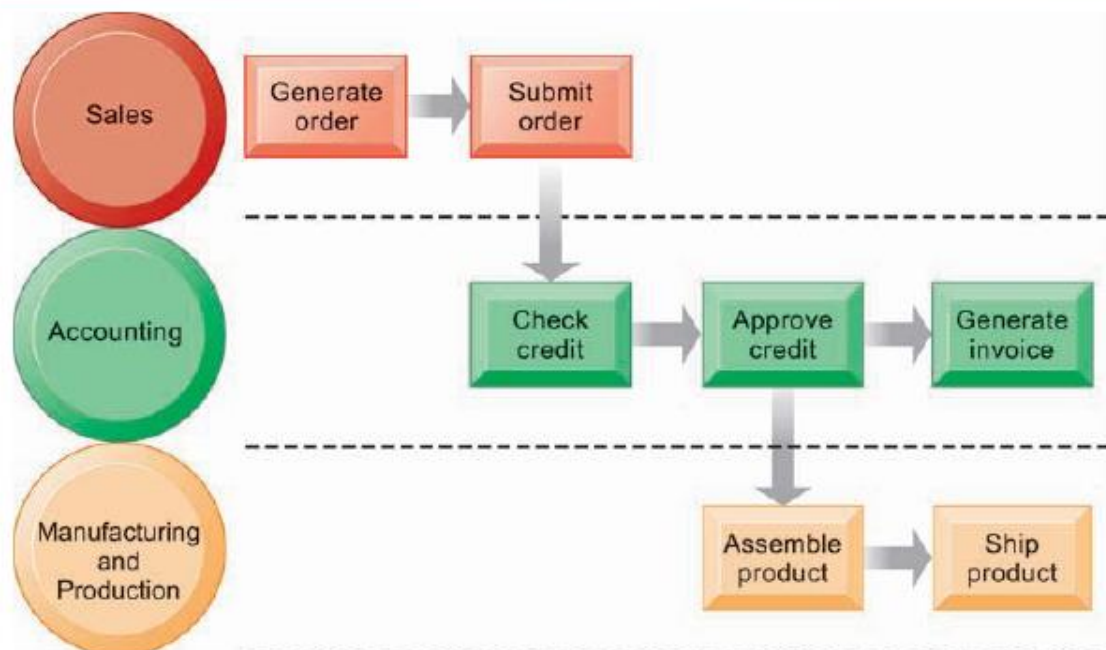
Course: Management Information System

Question:

How information systems could be used to support the order fulfillment process illustrated in the Figure below. What are the most important pieces of information these systems should capture? Explain your answer.

You need to write the source/ reference.

FIGURE 2.1 THE ORDER FULFILLMENT PROCESS



Fulfilling a customer order involves a complex set of steps that requires the close coordination of the sales, accounting, and manufacturing functions.

Supporting the Order Fulfillment Process through Information Systems

In a modern business, the order fulfillment process is more than just a sequence of tasks; it is a cross-functional journey that requires seamless coordination. As illustrated in the figure, this process bridges **Sales, Accounting, and Manufacturing**. Information Systems (IS)—specifically Enterprise Resource Planning (ERP) systems—act as the "digital nervous system" that allows these departments to talk to each other in real-time.

1. How Information Systems Support the Process

Information Systems transform this manual chain into an automated, efficient workflow in the following ways:

- **Eliminating Silos (Sales to Accounting):** When a salesperson "Submits an order," the system automatically alerts the Accounting department. There is no need for physical paperwork or manual emails. This ensures that the "Check credit" phase begins immediately, reducing wait times.
- **Automated Decision Making (Accounting):** Instead of a person manually digging through files, the IS can instantly compare the order value against the customer's remaining credit limit. It can "Approve credit" automatically based on pre-set rules, speeding up the transition to the next stage.
- **Triggering Action (Accounting to Manufacturing):** Once credit is approved, the system automatically sends a "Work Order" to the Manufacturing floor. The manufacturing team doesn't have to ask if the bill is paid; the system tells them it is time to "Assemble product."
- **Visibility and Tracking:** Throughout the process, the IS allows managers to see exactly where an order is. They can see if it's stuck in "Credit Check" or if it's currently being "Shipped," allowing for better customer service.

2. Critical Information Captured by the System

For this process to work effectively, the Information System must capture specific, high-quality data at each stage. The most important pieces of information include:

1. **Customer Identity & Profile:** (Name, ID, and contact details) – Vital for Sales and shipping.
2. **Order Details:** (Product SKUs, quantities, and special customizations) – Essential for Manufacturing to "Assemble" the correct item.
3. **Financial Data:** (Credit history, current balance, and payment terms) – Crucial for Accounting to "Approve credit."
4. **Inventory Status:** (Raw materials availability) – Needed before Manufacturing can commit to a timeline.
5. **Logistics Information:** (Shipping address and preferred carrier) – Necessary for the final "Ship product" step.
6. **Time Stamps:** (Date/time of order submission and approval) – Necessary to measure company efficiency (Lead time).

3. Summary Explanation

The beauty of using an Information System here is **synchronization**. Without IS, the Sales department might sell a product that Manufacturing doesn't have the parts to build, or Accounting might approve credit for a customer who hasn't paid their last three bills. By capturing and sharing data across these three departments, the system ensures that the business operates as one single unit rather than three separate islands.

Reference / Source

To ensure your assignment is academically valid, you can cite the following source, as this specific diagram is a foundational concept in their work:

Laudon, K. C., & Laudon, J. P. (2021). *Management Information Systems: Managing the Digital Firm* (17th ed.). Pearson.

This textbook is the primary source for the "Order Fulfillment Process" diagram used in most Business and IT courses worldwide. It explains how Enterprise Systems (ERP) integrate functional areas like Sales, Accounting, and Manufacturing.